Do Large Language Models Implicitly Encode Deontic Reasoning?

Exploring Faithfulness to Formal Deontic Logic

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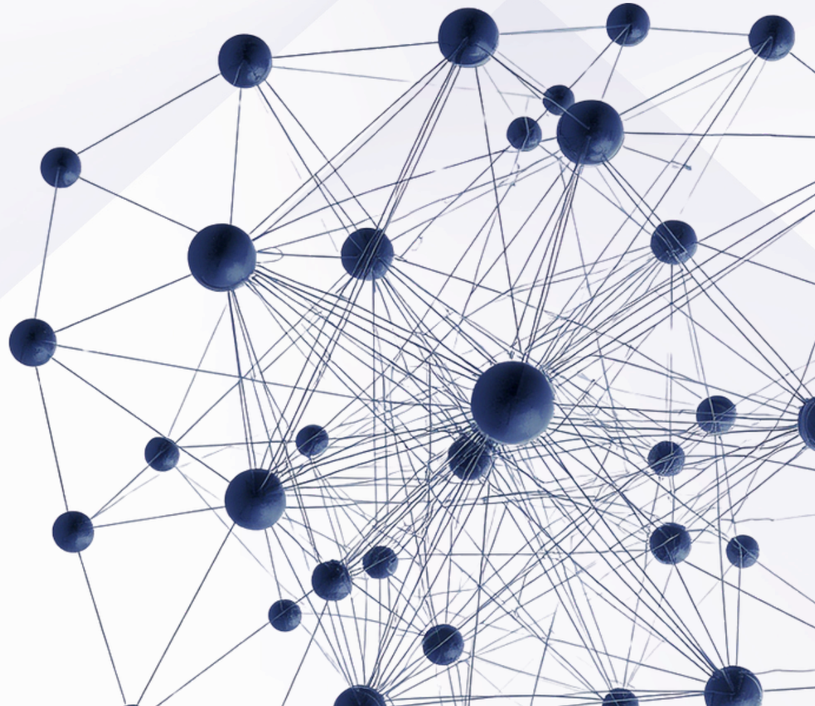


INTRODUCTION & SCOPE

Deontic logic studies obligations and permissions
("deon" = duty in Greek).

“You may leave" → true in some deontic worlds (permission)
"You must leave" → true in all deontic worlds (obligation).

Relevance to AI:
Do LLM consistenty exhibit rule usage, or dependent on form?



CORE TERMINOLOGY IN DEONTIC LOGIC

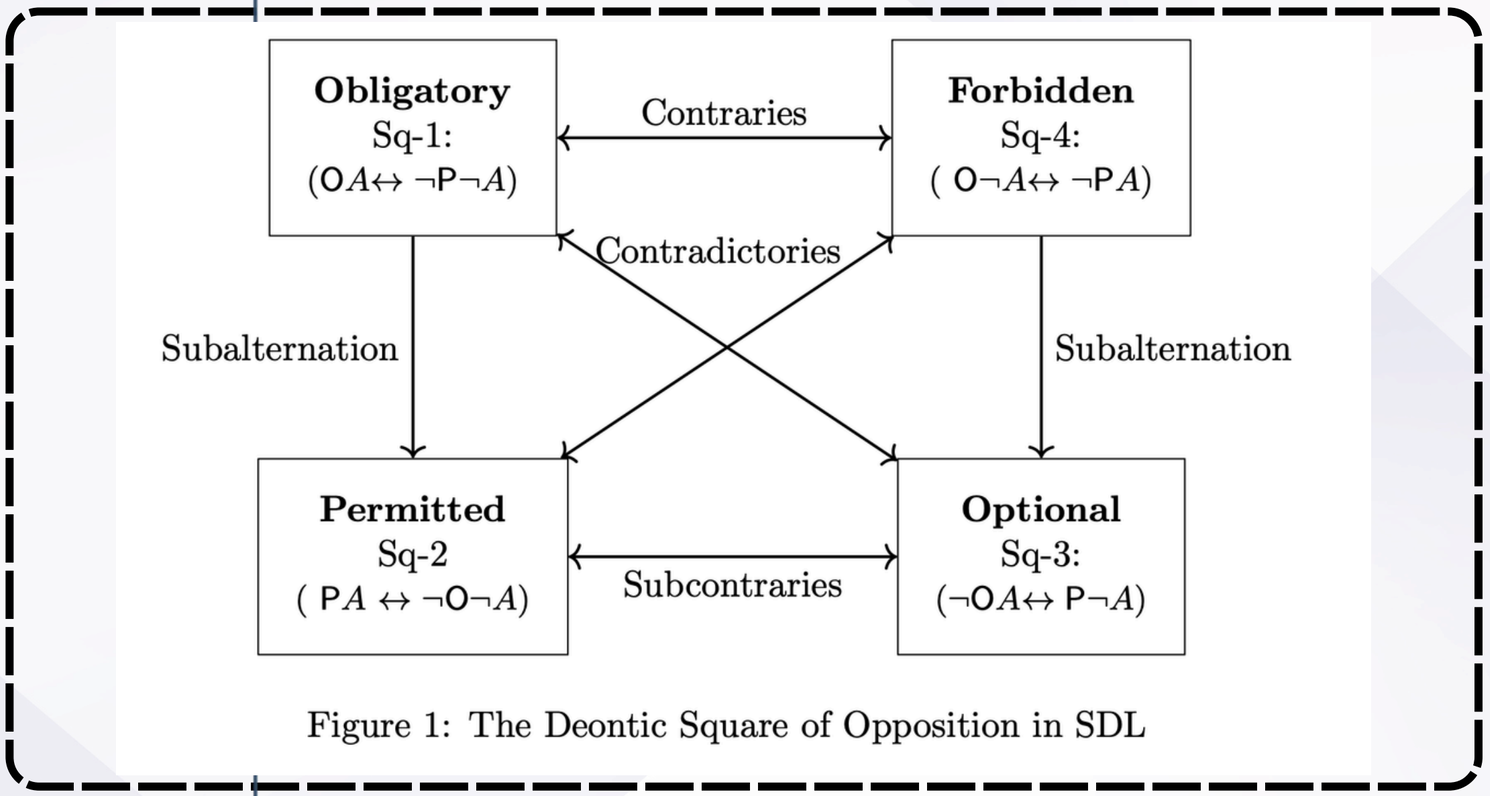


Figure 1: The Deontic Square of Opposition in SDL

RESEARCH QUESTIONS

RQ1: Which syntactic forms cause LLM failures in deontic reasoning tasks?

RQ2: Can LLMs recover semantic verdicts from explicit Kripke-style scenarios using formal structure rather than world knowledge?

RQ3: Which deontic proof rules do LLMs consistently follow or systematically violate?

RQ4: Are observed failure patterns consistent with known biases in language model reasoning?



BENCHMARK METHODOLOGY

GPT-4 | Claude | Deepseek V4 | Gemini

Models evaluated

Proof layer

Tests adherence to SDL axioms & deontic square

- **Inference rule tasks** — SDL axiom adherence
- **Deontic square** — O / P / F / PF relations
- **Failure patterns** — All 4 models analysed

Semantic layer

Eliminates world knowledge bias

- **Kripke scenarios** — Possible-worlds tasks
- **Nonsense predicates** — Strips statistical shortcuts
- **Formal vs. statistical** — Reasoning source probe

Syntax layer

Isolates syntactic processing capabilities

- **Controlled fragments** — Linguistic variation testing
- **Negation scope** — Boundary disambiguation
- **Operator inversions** — Templates & permutations

SDL AXIOMS

N (Necessitation): If $\vdash \varphi$, then $\vdash O\varphi$ — if something is a theorem, it is obligatory
K (Distribution): $O(\varphi \rightarrow \psi) \rightarrow (O\varphi \rightarrow O\psi)$ — obligation distributes over implication

D (Consistency): $OA \rightarrow PA$ — what is obligatory must be permissible
Definition: $PA \equiv \neg O\neg A$ — permission means the negation is not obligatory



KEY TEST RESULTS – SYNTACTIC AND SEMANTIC LAYERS

Paraphrase Consistency (S1): 0% failure rate

vocabulary variations and synonymous phrasing do not confuse models, demonstrating robust lexical flexibility in deontic reasoning tasks.

Negation Scope (S2): 0% failure rate

models reliably distinguish between absence of obligation versus explicit prohibition.

Operator Inversion (S3): Mostly correct responses;

notably, Deepseek V4 shows significant improvement when using Chain-of-Thought prompting.



KEY TEST RESULTS – CHALLENGING PARADOXES (S4 & S5)

Ross's Paradox & Free Choice (S4)
All models near-perfect. Models follow formal SDL logic even when it defies intuition.
$OA \rightarrow O(A \vee B)$
"If you must post the letter, must you post it or burn it?"

Contrary-to-Duty Paradoxes (S5)
Significant predicted failures. When obligations conflict, models prefer natural language coherence over formal consistency.
“you ought to keep your promise, but if you break it you ought to apologise.”



SEMANTIC LAYER TESTING WITH KRIPKE SCENARIOS

Testing formal semantic understanding through controlled scenarios:

- Nonsense predicates
- LLMs reason from formal logics OR on statistical patterns
- Kripke-style scenarios

’if the blewble was blopped, and the flig was zandered’ together with ’the flig was not zandered, can we infer ’the flig was not blopped’ ?

SUMMARY OF KEY FINDINGS

Models handle surface-level deontic language well
vocabulary changes and negation ("not allowed" vs. "forbidden") don't trip them up.

Negation is handled correctly
models reliably distinguish "not obligated" from "forbidden," with no failures recorded.

Whether deeper logic exposes further failures is still under investigation
semantic reasoning across Kripke scenarios and proof-layer consistency under chained obligations remain to be tested.

The pattern matching hypothesis is yet to be confirmed
whether models rely on statistical shortcuts rather than formal reasoning will be examined in the semantic and proof layers.



IMPLICATIONS, LIMITATIONS, AND FUTURE DIRECTIONS

Implications:
• Deontic reasoning is crucial for AI systems handling norms, rules, and ethics
• Mixed capabilities in current LLMs highlight significant gaps in formal logical understanding
• Models may produce plausible-sounding but logically inconsistent normative judgments

Limitations & Future Directions:
• Rapidly evolving LLM landscape may change results over time
• Difficulty isolating effects of architecture vs training vs prompt engineering
• Future work: mechanistic analyses of failure causes and continuous evaluation as models develop

THANK YOU

Questions and Discussion
Welcome